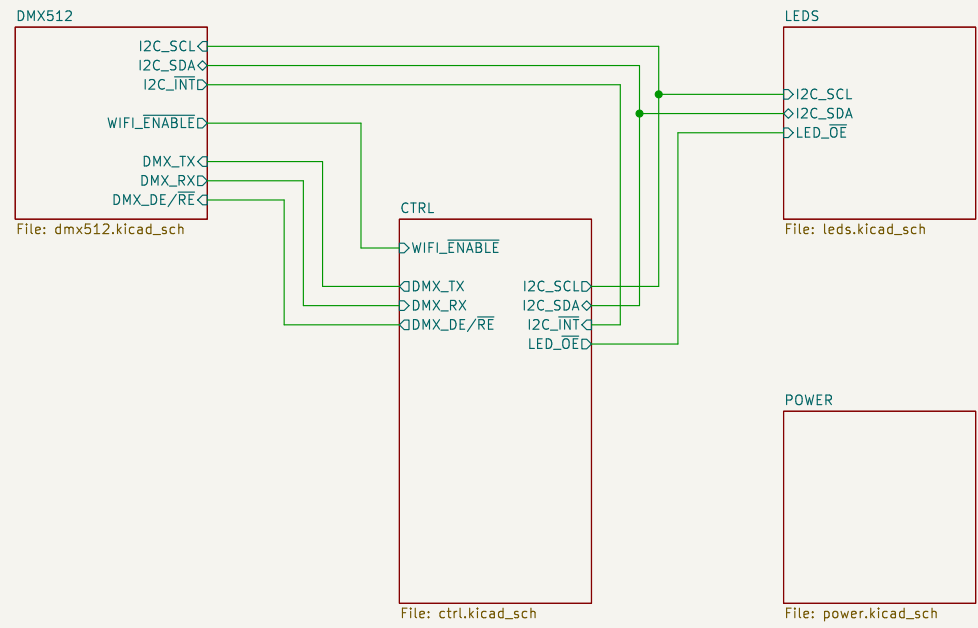
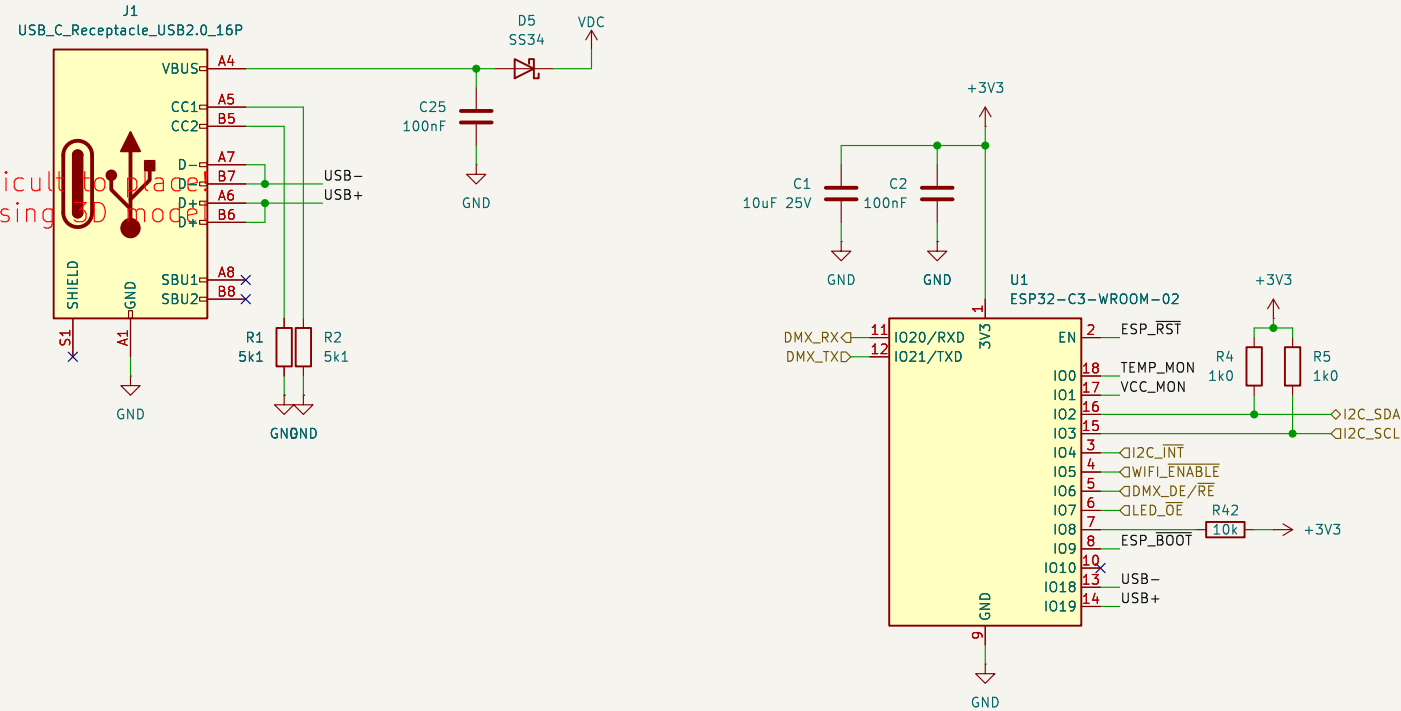


LED Beacon



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

[illegible]

Auto reset

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF

C27
25V

SW2
SW_Push

GND

GND

GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Auto reset

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF

C27
25V

SW2
SW_Push

GND

GND

GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Auto reset

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF

C27
25V

SW2
SW_Push

GND

GND

GND

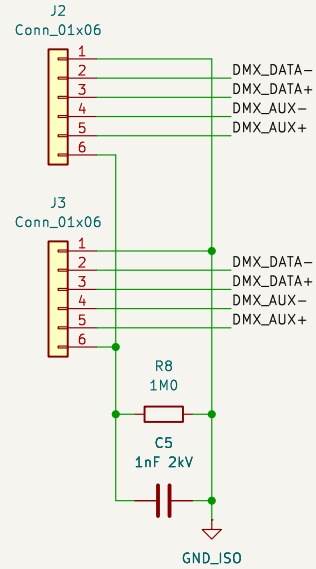
Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LEDs?

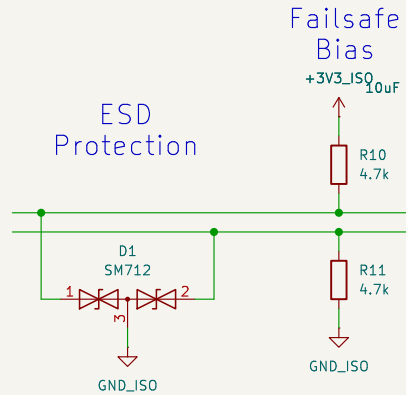
Sheet: /CTRL/ File: ctrl.kicad_sch	
Title:	
Size: A4	Date: 2026-04-10
KiCad E.D.A. 9.0.9	Rev: 0 Id: 2/5

DMX512

To XLRs



ESD Protection



Failsafe Bias

+3V3_ISO

10uF 25V

C7

4.7k

R10

GND_ISO

DMX_DATA+

DMX_DATA-

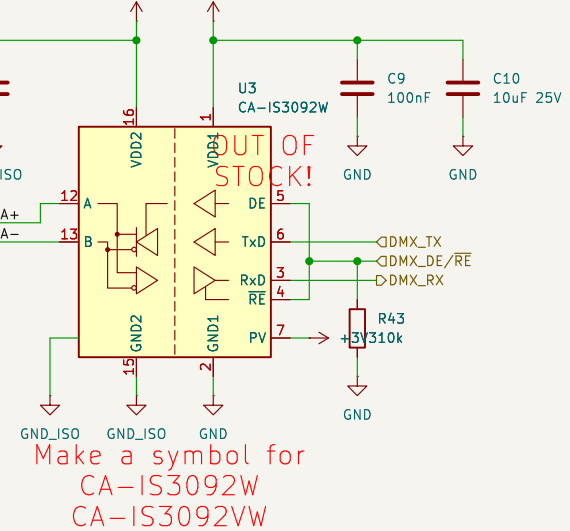
R11

4.7k

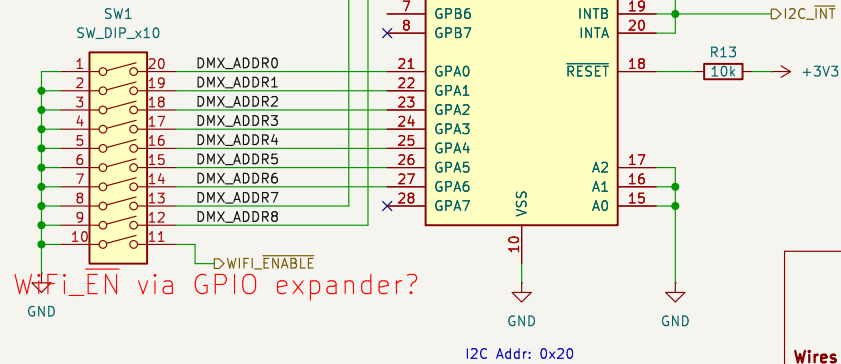
GND_ISO

Isolated RS485 TRX

+3V3_ISO +3V3



GPA7 & GPB2:7 for?



https://dfimg.drobot.com/wiki/23269/DFR0944_esp32-s3-based-dmx512-smart-lighting-controller_schematics_V1.0.pdf

<https://www.ti.com/lit/an/slla272d/slla272d.pdf>

Wires & Bits

Sheet: /DMX512/

File: dmx512.kicad_sch

Title:

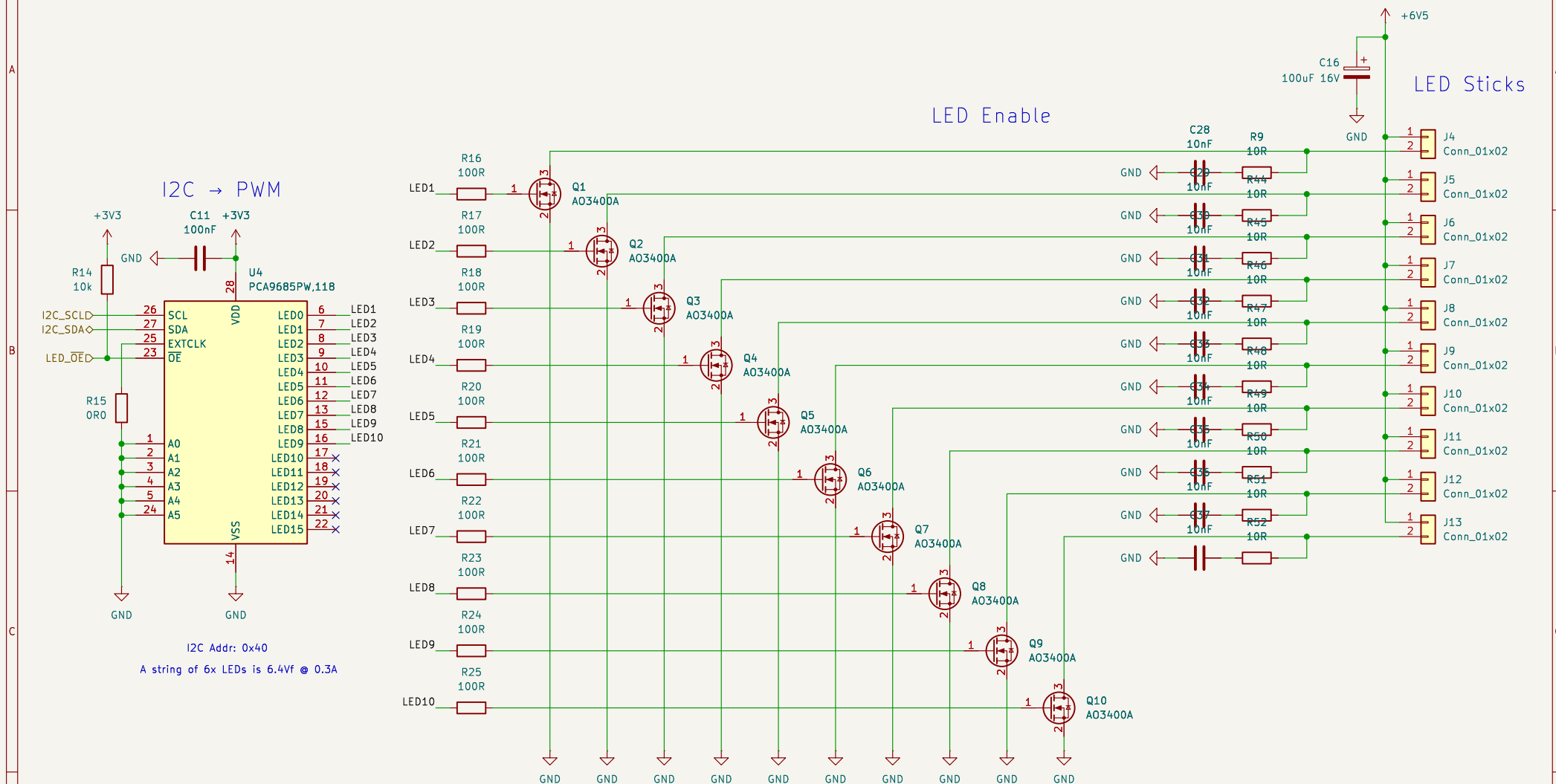
Size: A4 Date: 2026-04-10

KiCad E.D.A. 9.0.9

Rev: 0

Id: 3/5

LED Driver



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

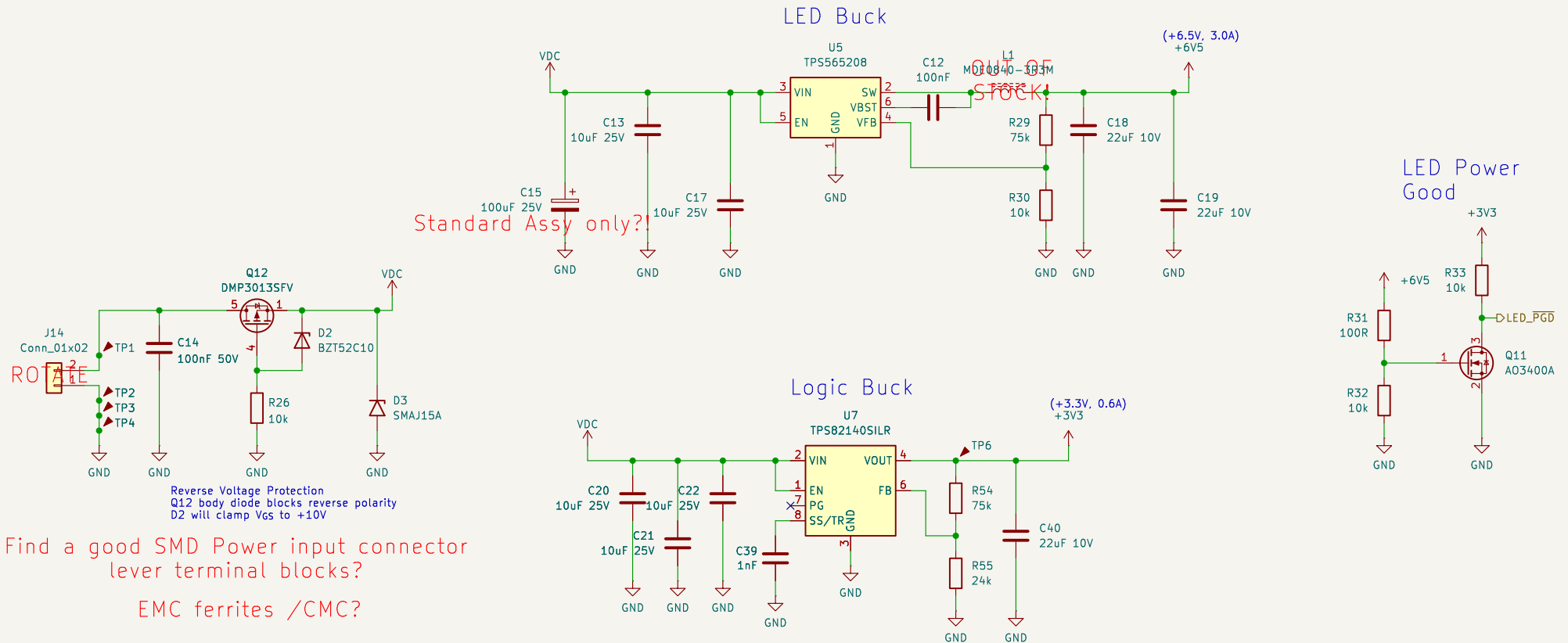
Title:

Size: A4	
KiCad E.D.A. 9.0.9	

Date: 2026-04-10

Rev: 0
Id: 4/5

Power Supplies



Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4

Date: 2026-04-10

KiCad E.D.A. 9.0.9

Rev: 0

Id: 5/5